

RESEARCH ARTICLE

Mission Profile-Based Reliability Assessment for Modular MVAC-LVDC Solid-State Transformers

AHMED MELIGY^{1,2}, RAFAEL COELHO-MEDEIROS¹,
ILKNUR COLAK¹, (Senior Member, IEEE), AND SEDDIK BACHA², (Senior Member, IEEE)

¹Schneider Electric, 38000 Grenoble, France

²Univ. Grenoble Alpes, CNRS, Grenoble INP, G2ELab, 38000 Grenoble, France

Corresponding author: Ahmed Meligy (ahmed.meligy@se.com)

ABSTRACT This paper introduces a comprehensive framework for evaluating the reliability of modular MVAC-LVDC Solid-State Transformers (SSTs) based on their mission profiles. The proposed approach emphasizes the importance of accurately assessing the reliability of Medium-Frequency Transformers (MFT), which play a critical role in SST performance. The study presents: 1) a mission profile-based methodology for analyzing MFT reliability; 2) a detailed process for estimating SST lifetime that accounts for both random and wear-out failure mechanisms; and 3) an exploration of how MFT reliability influences overall SST lifespan. Additionally, the paper outlines the design methodology for MVAC-LVDC SSTs and details the component selection process used in the reliability assessment. Two case studies—one involving an electric vehicle load and the other a data center load—are applied to a 6.32 MVA, 11 kV AC / 1 kV DC SST design. Results indicate that incorporating MFT reliability into the analysis may reduce the estimated SST lifespan by over 15%, highlighting the value of integrated reliability assessments in SST design. This assessment identifies components prone to failure, allowing for redesigns that improve resilience against mission profile stresses and extend the overall converter lifetime.

INDEX TERMS Electro-thermal modeling, medium-frequency transformers, rainflow counting, reliability, solid-state transformer, wear out.

I. INTRODUCTION

The Solid-State Transformer (SST) is a relatively high-power density conversion solution designed to interconnect electrical systems operating at different voltage levels—High Voltage (HV), Medium Voltage (MV), and Low Voltage (LV)—and of different nature, whether Direct Current (DC) or Alternating Current (AC). Driven by the growing demand for LVDC interconnections to distributed sources and dense loads, SSTs have emerged as a viable alternative for conventional Low-Frequency Transformers (LFTs) and power factor correctors. SSTs integrate electrical isolation, voltage transformation, and reactive compensation functions through power electronic circuitry alongside one or more Medium Frequency Transformers (MFTs), while ensuring backward compatibility with existing grid infrastructure.

The associate editor coordinating the review of this manuscript and approving it for publication was Truong-Duy Duong¹.

Their development focuses on enhancing power density, controllability, flexibility, and efficiency in electrical systems, making them suitable across diverse domains: railway traction, transmission and distribution grids, microgrids, solar and wind power generation networks, electric vehicle charging, data center power provisioning, shipboard electrical networks, and energy storage systems [1].

The SST comprises multiple fundamental conversion units tailored to the interconnected systems, with each additional conversion stage offering an extra degree of freedom for enhanced control capabilities [1]. The SSTs are therefore categorized based on their conversion stages and modularity, where each topology presents distinct advantages and trade-offs. In AC-DC applications, including MVAC-LVDC, SSTs have been validated as economically viable alternatives to conventional systems [2]. These can employ either single-stage or two-stage configurations, with the latter enabling input-output decoupling through DC links at both

MV and LV levels [3]. To effectively meet the voltage and power requirements of MV applications, a modular Input Series Output Parallel (ISOP) arrangement of converter modules is often employed. This approach facilitates voltage and current sharing among cascaded cells, thereby enhancing scalability and improving fault-resilience through redundancy. However, the large number of power electronic devices in modular designs introduce reliability concerns. As all components are required to operate simultaneously, the likelihood of failure increases, creating key technical bottlenecks that limit SST deployments. Although adding redundant modules can enhance reliability, it may also result in considerably higher costs. Therefore, finding the right balance between modularity and reliability is vital in SST design. To be a viable alternative to conventional systems, the longevity of SSTs should meet or surpass the already high standards set by their competition, while ensuring comparable or improved efficiency.

While reliability studies on switching devices and capacitors are well-established [4], [5], [6], [7], [8], [9], [10], [11], [12], research on the reliability and lifespan assessment of MFT remains largely underexplored in the literature. Previous works on SST reliability often overlook the influence of MFT lifespan on overall system reliability. For example, in [13], the authors examine the cost and reliability of different SST module topologies, but their assessment relies on constant failure rates and focuses solely on semiconductor devices. In [14] and [15], the authors combine reliability evaluation with redundancy design for a modular DC/DC SST focusing on wear-out failure analysis, yet they omit MFTs from their evaluation. Similarly, in [16], a mission profile-based reliability assessment for SSTs in railway applications is presented, but it also excludes MFTs. In [3] and [17], the authors propose a power routing method between SST modules to enhance the system's lifespan, but their analysis estimates the remaining useful lifetime of the modules solely through power semiconductors. Likewise, in [18], a power routing control method based on virtual resistance is presented. While both power switches and capacitors are investigated in this work, the MFT reliability is yet omitted from the analysis. Conversely, the authors in [19] introduce a reliability-oriented SST design that employs a second harmonic current routing method, incorporating MFT reliability into their assessment, however they use empirical lifetime equations for LFTs to estimate MFT lifespan. In [20] and [21], the authors conducted a comparative analysis of various topology variations for cascaded modular MVAC-LVDC SSTs, evaluating multiple criteria, including reliability. The findings indicate that the front-end half-bridge matrix-based dual active bridge module offers the best compromise among the criteria studied. However, the reliability assessment was based solely on the random failure analysis of the semiconductors and capacitors. Moreover, reliability assessments of single-stage isolated converters similarly either overlook MFT reliability altogether [22], [23], [24], [25] or rely on empirical equations from reliability

handbooks that provide constant failure rates to estimate the MFT lifespan [26], [27], [28], [29], [30].

Prior research has made strides in understanding SST reliability; however, a research gap remains in adequately addressing the lifetime of MFTs and their impact on the overall reliability of SSTs. When the SST is connected to medium-voltage (MV) mains, it is essential for the MFT to provide, at a minimum, medium-voltage functional isolation in accordance with the environmental pollution degree outlined in IEC 60664-1 [31] to ensure operational safety. In addition to complying with MV insulation standards, MFTs must withstand electric field stresses caused by pulse-width modulation (PWM) voltage during both normal and transient conditions. The combination of high voltage and high-frequency waveforms, along with the thermal challenges stemming from the compact design of MFTs, can lead to increased Partial Discharge (PD) activity within the insulation. This rise in PD activity increases the risk of dielectric material breakdown within the MFT and accelerates aging [32], [33], [34], [35], potentially reducing the lifetime of the SST.

Given these challenges, an SST reliability study considering the lifetime of MFTs is crucial for understanding the remaining lifespan of the entire system. Future research should aim to create comprehensive reliability models that take into account the unique characteristics and failure modes of MFTs. Therefore, this paper contributes to three key areas: firstly, it proposes a mission profile based theoretical framework for estimating MFT reliability within MVAC/LVDC SSTs; secondly, it provides the complete methodology for lifetime estimation of the SST that incorporates both random failures derived from reliability handbooks and wear-out failures assessed through a cycle-counting approach; and thirdly, it demonstrates the effects of MFT reliability when factored in the SST lifetime estimation. As system reliability is integral to the design process, this paper also briefly outlines the design methodology for an MVAC-LVDC SST and details the component selection for which the mission-based reliability assessment is performed.

The remainder of this paper is structured as follows. Section II provides a brief overview of reliability metrics. Section III describes the SST design and components selection used in this work. Section IV illustrates the power losses within the SST components and their translation to electro-thermal models. Section V describes each of the components reliability estimation procedure. Section VI provides results of different test cases for the SST reliability model. Section VII summarizes the findings of this study.

II. RELIABILITY METRICS: A BRIEF OVERVIEW

Reliability is defined as the probability that a device will not fail for a specified duration under nominal operating conditions. In modern power electronics based systems, understanding the expected lifespan of converters is essential for making informed decisions regarding facility planning, cost-effective design, and replacement scheduling [36].

There are five critical reliability metrics that are used to assess the lifetime of a device [37], [38], [39]. Firstly, the failure probability density function (*pdf*) denoted by $f(t)$ which indicates the failure distribution over a device's entire lifetime. Secondly, the failure cumulative distribution function (*cdf*), also called the unreliability function and is denoted $F(t)$, which describes the probability of failure by a specified time t . Thirdly, the reliability function $R(t)$, which is the complement of $F(t)$, and represents the probability of the device operating without failure between time 0 and time t , with $R(0) = 1$ and $R(\infty) = 0$.

$$R(t) = 1 - F(t) = 1 - \int_0^t f(x)dx \quad (1)$$

The fourth metric is the hazard/failure rate $h(t)$ which represents the likelihood of a device failing in the upcoming time interval, provided that it has been functioning correctly up to time t , essentially indicating the instantaneous risk of failure.

$$h(t) = \frac{f(t)}{R(t)} \quad (2)$$

Finally, the Mean Time To Failure (MTTF) which signifies the equivalent lifetime in operation and is calculated as

$$MTTF = \int_0^{\infty} R(t)dt \quad (3)$$

Fig. 1 depicts the standard hazard function for a device over its lifecycle, highlighting three distinct phases: early failures, useful life, and wear-out failures [36], [37], [38], [39], [40]. Early failures are often overlooked in reliability studies, as they generally arise from manufacturing defects or debugging issues, which are typically resolved prior to the device being put into operation. During the useful lifetime the device may experience random failures due to unexpected internal faults or external environmental factors. On the other hand, as materials age, the device may enter the wear-out phase, where failure rates increase due to the effects of material degradation and the cumulative stresses experienced during operation.

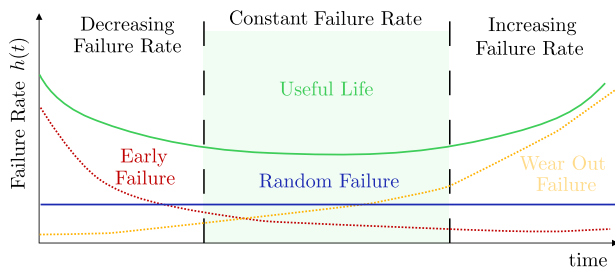


FIGURE 1. Failure rate bathtub curve.

In general, a component or system fails when a specific failure mechanism is triggered, either by random chance or by wear-out processes. The total failure rate of the device, denoted as $h_{tot}(t)$, is the sum of the random failure rate ($h_r(t)$) and the wear-out failure rate ($h_{wo}(t)$). When assessing

a system of multiple components, if the failure of any individual component leads to the failure of the entire system, a series reliability network is utilized where the system's failure rate is the sum of all individual components failure rates.

$$h_{sys,tot}(t) = \sum h_{comp,r}(t) + \sum h_{comp,wo}(t) \quad (4)$$

Conversely, systems with redundant structures and standby components utilize a combination of parallel and series reliability networks, for which reliability techniques like Markov models are used [38], [39].

A. RANDOM FAILURE

Predicting the failure rate during the useful lifetime of a device primarily involves analyzing historical failure data from previous operations, with more accurate data obtained from long-term use under consistent operating conditions. Reliability handbooks, such as the widely recognized Military Handbook 217 (MIL-HDBK217) [41], along with others like Telcordia SR-322, Siemens SN29500, RDF-2000, IEC-TR-62380, IEC-61709, and FIDES, provide generic estimates of failure rates for various electronic components based on available operational data. These handbooks offer a base failure rate (λ_b) for each component and incorporate various stress factors (e.g., electric factor - π_S , temperature stress - π_T , quality factor - π_Q , environmental factor - π_E , ..., etc.) to account for their effects on the component's reliability. Furthermore, one can utilize data supplied by manufacturers or users as the foundational failure rate to estimate the failure rate for specific operating conditions.

Random failure is often modeled using the exponential distribution function, as presented in (5) [38], [39]. The random failure rate ($\bar{\lambda}_r$), which for the exponential reliability function is equivalent to $h_{comp,r}(t)$, represents the mean of the instantaneous failure rate calculated at each operating point i across n total points.

$$R(t) = e^{-\bar{\lambda}_r t} \quad (5)$$

$$\bar{\lambda}_r = \frac{1}{N} \sum_{i=1}^N \lambda_b^{S,T,Q,E,...} \prod_k \pi_k \quad (6)$$

Although these handbooks are still used for lifetime estimations, some of them have several notable concerns, including outdated data, unclear failure mechanisms, and some exclude various operating conditions [36]. Therefore, the assumption of a constant failure rate in these handbooks can result in unrealistic projections, especially in cases where components enter the degradation phase during their mission profile, in which case wear-out failure must be considered.

B. WEAR-OUT FAILURE

Wear out failure analysis is based on a Physics-of-Failure (PoF) approach, which emphasizes the investigation of root-cause failure mechanisms and how materials, defects, and stresses affect product reliability. This method distinguishes between overstress failures, which result from

a single event, and wear-out failures, which arise from cumulative damage due to the mission profile. Unlike empirical failure analysis, which relies on historical data, the PoF approach is rooted in deterministic science, necessitating an understanding of component material properties, failure mechanisms, and mission profiles [40]. As a result, degradation cannot be generalized across different mission profiles without conducting extensive tests on multiple devices to establish a failure probability density function. Therefore, lifetime estimates are generated from accelerated aging tests, enabling failure rate predictions for similar devices that share technology and operating conditions [42].

A statistical method can be employed to account for potential variations in the calculated measurements and parameters of the lifetime equation using Monte Carlo simulations [40]. The lifetime data derived from the Monte Carlo analysis is then commonly fitted to a Weibull probability density function (*pdf*), as described in (7), where α and β indicate the scale and shape parameters, respectively.

$$f(t) = \frac{\beta}{\alpha} \left(\frac{t}{\alpha}\right)^{\beta-1} e^{-\left(\frac{t}{\alpha}\right)^{\beta}} \quad (7)$$

III. SST DESIGN

While the concept of the SST presents manifold advantages, its design and realization present significant technical challenges, particularly in topology selection and reliability optimization. SSTs can be broadly classified based on their modularity into monolithic, modular, and semi-modular architectures [1], [43], [44], [45], [46]. Monolithic configurations use a single power converter rated for full power, requiring HV power switches or a series association of LV power switches to scale up for the MV applications. While this reduces the number of components and simplifies control strategies, it may compromise fault tolerance and scalability. Modular designs consist of multiple converter modules, with the ISOP configuration being the most common for MV/LV conversions. This approach offers voltage/current scalability, redundancy, and reduced electromagnetic interference

emissions. Semi-modular configurations adopt a hybrid approach, often combining a modular MV-side converter with a single LV-side converter via an MFT. This approach reduces the component count while retaining some advantages of modularity.

A wide variety of SST topologies have been proposed by both academia and industry for which [20], [21], [43], [44], [45], [46] provide a comprehensive summary. These generally comprise three fundamental conversion units—AC/DC, AC/AC, and DC/AC—with an MFT providing galvanic isolation. These building blocks allow SST architectures to be categorized based on the number and role of DC links. Single-stage and quasi-single-stage topologies typically include a single DC link at the LV output, implemented through direct and indirect matrix converters, respectively, at the front end of the SST. In contrast, two-stage converter architectures employ two decoupled DC links: one at the MV side and one at the LV output. This decoupling enables independent control of the upstream and downstream converters, providing more control flexibility.

It is important to recognize that every aspect of the SST design process influences the overall converter lifetime. A major source of reliability variation stems from the chosen architecture, whether single-stage, quasi-single-stage, or two-stage, as well as the specific converter topology employed at each stage. These configurations differ in component count, operating principles, modulation strategies, use of resonant circuitry, and switching frequencies, all of which affect stress distribution and lifetime estimation. Furthermore, the mission profile and the stress tolerance of individual components significantly impact system longevity. Lower overall efficiency can also lead to increased thermal losses, exposing components to elevated temperatures, thereby accelerating aging mechanisms.

Among the various SST topologies reported in the literature, one of the most widely adopted configurations is the modular two-stage conversion architecture, in which each SST module employs a Full-Bridge Voltage Source Converter (FB-VSC) at both the MV and LV stages, as illustrated in

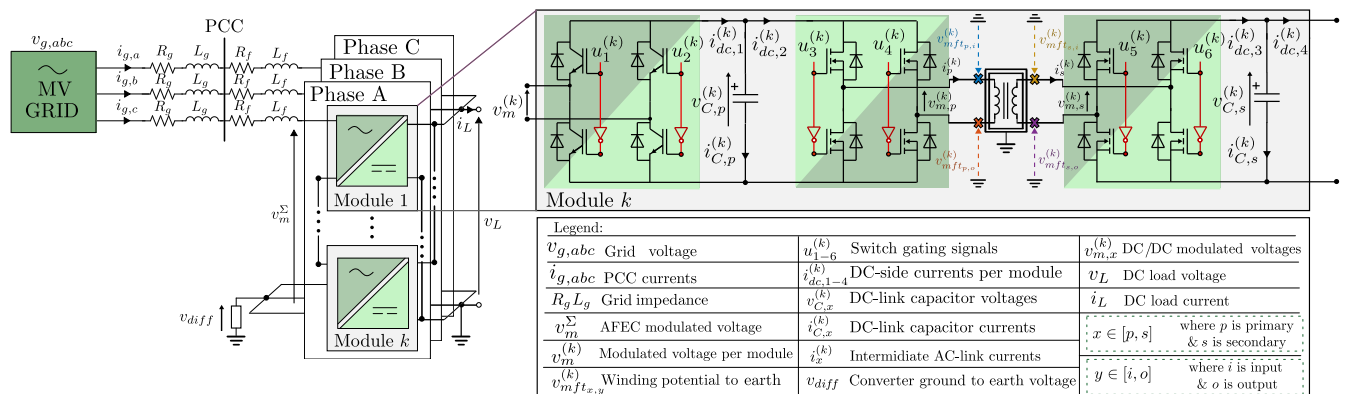


FIGURE 2. MVAC-LVDC ISOP SST.

Fig. 2. This topology was selected for the present study due to its relative simplicity and reduced component count compared to more complex alternatives. While the reliability assessment framework developed in this work is applicable to any SST topology or architecture, this particular configuration was chosen as a representative baseline to establish and demonstrate the methodology. It serves as a foundation for future investigations into the reliability performance of more advanced solid-state transformer designs.

A. REFERENCE CASE

In this study we consider a 6.32 MVA, 11 kV AC / 1 kV DC SST which schematic is illustrated in Fig. 2, and has the fixed parameters presented in Table 1 for which the components selection is detailed in the following subsections. The MVAC grid is modelled by an equivalent three phase source $v_{ga,b,c}$ in series with its equivalent impedance L_g , R_g . In the remainder of this article, the phase indices (a, b, and c) are omitted from the variables for simplicity. The ISOP SST is connected to the grid through a line inductor with inductance L_f and resistance R_f . The grid current and modulated voltage are denoted i_g and v_m^Σ , respectively. Whereas the current and voltage at the LVDC side are denoted i_L and v_L , respectively. The Active Front End Converter (AFEC) connects the AC grid to the primary-side DC bus $v_{C,p}^{(k)}$ of the Isolated DC/DC converter. In addition, it employs Phase-Shifted PWM (PS-PWM) to generate the modulated voltage $v_m^{(k)}$, where $k \in [1, \dots, N^{mod}]$, and N^{mod} represents the total number of modules per phase. The isolated DC/DC converter operates as a Dual-Active Bridge (DAB), utilizing trapezoidal modulation and featuring a medium frequency link that consists solely of the MFT to maintain low leakage inductance for high efficiency conversions [47].

B. COMPONENTS SELECTION

To attain high conversion efficiency in the DAB stage, the primary DC-link voltage is configured to match the secondary DC-link voltage, ensuring a primary-to-secondary DC voltage ratio ($N_t V_{dc,s} / V_{dc,p}$) of 1 with a 1:1 turns ratio in the MFT. This DC voltage ratio is established to enable trapezoidal modulation throughout the full operational

range [47]. The equal turns on the primary and secondary windings simplify the design, enhance magnetic coupling, and reduce leakage inductance, which minimizes core and copper losses thereby increasing the efficiency of the MFT.

With 1 kV DC-links as per the system requirements and utilizing a FB-VSC, the maximum modulated Root Mean Square (RMS) voltage ($V_{m,rms}^{(k)}$) for a single module is calculated to be 0.7071 kV. Whereas, the RMS maximum modulated AC grid voltage requirement $V_{m,rms}^\Sigma$ is calculated as

$$V_{m,rms}^\Sigma = \sqrt{\text{Re}(\bar{v}_{m,rms}^\Sigma)^2 + \text{Im}(\bar{v}_{m,rms}^\Sigma)^2} \quad (8)$$

where,

$$\bar{v}_{m,rms}^\Sigma = \frac{\bar{v}_{gll} \cdot k_g}{\sqrt{3}} + (R_{tot} + jX_{tot})\bar{i}_g \quad (9)$$

whereas, $R_{tot} = R_g + R_f$, $X_{tot} = 2\pi f_0(L_g + L_f)$, k_g denotes the maximum grid overvoltage limit, $\bar{v}_{gll}$ is the complex line-to-line grid RMS voltage, and $\bar{i}_g$ represents the complex RMS grid current at rated power. The real and imaginary part of $\bar{v}_{m,rms}^\Sigma$ can be derived as

$$\begin{aligned} \text{Re}(\bar{v}_{m,rms}^\Sigma) &= \frac{\text{Re}(\bar{v}_{gll}) \cdot k_g}{\sqrt{3}} + R_{tot} \cdot \text{Re}(\bar{i}_g) - X_{tot} \cdot \text{Im}(\bar{i}_g) \\ \text{Im}(\bar{v}_{m,rms}^\Sigma) &= \frac{\text{Im}(\bar{v}_{gll}) \cdot k_g}{\sqrt{3}} + R_{tot} \cdot \text{Im}(\bar{i}_g) + X_{tot} \cdot \text{Re}(\bar{i}_g) \end{aligned} \quad (10)$$

The minimum number of modules required per phase can then be computed as follows and used to determine the optimal blocking voltage of the deployed semiconductors [48],

$$N_{min}^{mod} = \text{ceil}\left(\frac{V_{m,rms}^\Sigma}{V_{m,rms}^{(k)} \cdot M_{max}}\right) \quad (11)$$

where, M_{max} is the maximum modulation index of the converter. For this analysis, we have assumed an upper grid overvoltage limit of 1.15 p.u. and set M_{rms} to 0.95. This results in $N^{mod}=12$ modules per phase, each rated at 175.7 kVA for an 11 kV line-to-line grid. The switching frequency for the AFEC ($f_{sw,AFE}$) using PS-PWM, is established at 250 Hz to minimize modulation harmonics within the 50th-order window, ensuring that the first sideband harmonics occur at the $2N^{mod} f_{sw,AFE}$ order. Consequently, 1.7 kV Si-IGBT switches are utilized, given their compatibility with the low switching frequency and the defined DC bus voltage. The DAB converter operates at a higher switching frequency ($f_{sw,DC} = 15$ kHz) to reduce the weight and volume of the MFT. In this scenario, 2 kV SiC MOSFETs are considered owing to their low switching losses at high frequencies and reduced heat generation. The specifications of the selected switches for this study are provided in Table 2.

The primary DC-bus capacitor must filter the current harmonics generated by the AFEC and the primary side of the DC/DC converter. The DC/DC converter produces high-frequency harmonics at multiples of its switching

TABLE 1. SST main system parameters.

Network Parameter	Value	Units
Grid ph-ph RMS Voltage (V_{gll})	11	kV
Nominal Apparent Power	6.32e6	VA
Nominal Active Power (P_{nom})	6e6	W
Fundamental Frequency (f_0)	50	Hz
Grid Resistance (R_g)	58.1e-3	Ω
Grid Inductance (L_g)	7.48e-4	H
Line Filter Resistance (R_f)	1e-3	Ω
Line Filter Inductance (L_f)	15e-3	H
Number of Modules per phase (N)	12	-
Primary Capacitance (C_p)	6e-3	F
Secondary Capacitance (C_s)	5.6e-4	F
AFEC switching frequency ($f_{sw,AFE}$)	250	Hz
Isolated DC/DC switching frequency ($f_{sw,DC}$)	15e3	Hz
Nominal DC-link voltages ($V_{dc,p}, V_{dc,s}$)	1	kV

TABLE 2. Module components & properties [49], [50], [51].

Part	Component	Reference	Properties	Amount
AFEC	Si-IGBT	FF450R17ME4_B11	1.7 kV, 450 A	4
DAB	SiC MOSFET	FF3MR20KM1H	2 kV, 400 A	8
Primary DC-link	Film cap.	B25620B0408K703	0.7 kV, 100 A	2 series 3 parallel
Secondary DC-link	Film cap.	B25620B0567K703	0.7 kV, 80 A	2 series 2 parallel

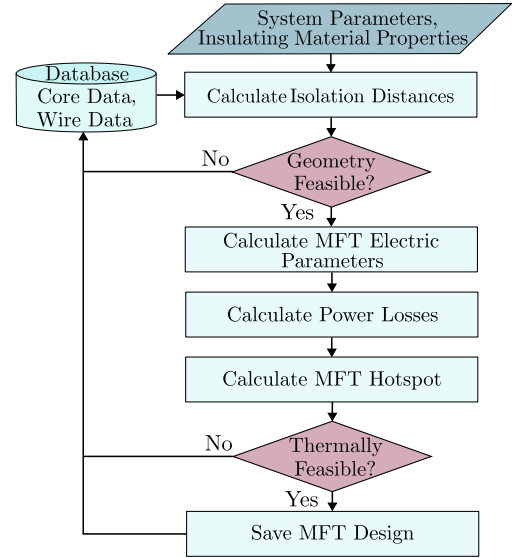
frequency of 15 kHz, while the AFEC generates harmonics starting from double the fundamental frequency. Lower-frequency harmonics encounter significantly higher impedance than higher-frequency ones, which leads to increased losses and voltage ripple thereby increasing the capacitor filter requirements on the primary DC-bus. While the low-order current harmonics, at integer multiples of the grid frequency, are transferred from the primary to the secondary side, they are, however, attenuated by the MFT inductance. Consequently, the current in the secondary DC-link is characterized by even multiples of the DC/DC converter's switching frequency, along with sidebands at integer multiples of the fundamental frequency.

The minimum required capacitance for each of the DC-links is calculated based on the maximum charge variation corresponding to the most demanding grid operating point, while ensuring a maximum voltage ripple of 5%. This point corresponds to the lower operational bound of the grid voltage at rated apparent power transfer. The specifications for the selected capacitors are presented in Table 2.

C. MFT DESIGN

While switches and capacitors are off-the-shelf products, high-power MFTs are custom-made based on the system's ratings. In this study, we follow a design procedure similar to those presented in [52], [53], and [54] to design a 15 kHz, 170 kW MFT. The flowchart of the MFT design procedure is presented in Fig. 3.

The design follows an iterative process in which the algorithm loops through a database of different cores, along with their corresponding material properties and wire types. It takes as input the electrical system parameters (P_{nom} , $V_{dc,(p,s)}$, f_{sw} , N_t) as well as the insulation material properties of the windings and wires. An initial MFT sizing is determined from these input parameters and checked for feasibility in terms of geometry. If the design is dimensionally feasible, the insulation distances are calculated and compared to the minimum requirements specified by the Basic Insulation Level (BIL) standards. According to the IEC 60664-1 standards, a medium-voltage system operating at 11 kV must typically have a standard BIL rating of 60 kV [31]. The insulation distances based on the material are calculated by (12), where V_{di-ins} represents the maximum potential difference the insulation must withstand, E_{di-ins} is the frequency-dependent dielectric strength of the insulating

**FIGURE 3. MFT design algorithm.**

material, and k_{saf} is an additional safety margin that accounts for transient overvoltage, and is set to 0.3 in this study.

$$d_{iso} = \frac{V_{di-ins}}{k_{saf} E_{di-ins}} \quad (12)$$

To determine the different V_{di-ins} within the MFT, the potential of each wire turn in the MFT to ground must be calculated. This is done by first computing the voltage at the input and output terminals of both the primary and secondary windings, these reference points are marked by crosses in Fig. 2. By tracing the path from each of these reference points to ground, a generalized equation can be derived for the potential difference of each of the four MFT terminals to ground. These equations are formulated in the frequency domain as presented in (13)-(16), where $V_{u,(1...6)}$ denotes the voltage across the upper switches of the corresponding switching cell controlled by the switching signals u_1 through u_6 . The time-domain waveforms for Module 1, corresponding to these equations, are shown in Fig. 4. Once the terminal voltages are known, the voltage on each individual wire turn within the MFT can be determined by assuming a constant voltage drop between adjacent conductors, ensuring a uniform voltage gradient throughout the winding structure, based on the spatial positioning of each conductor.

$$V_{mft_{p,i}}^{(k)} = \sum_n^{k:N^{mod}} V_m^{(n)} + V_{u,1}^{(n)} - V_{u,3}^{(n)} + V_{diff} \quad (13)$$

$$V_{mft_{p,o}}^{(k)} = \sum_n^{k:N^{mod}} V_m^{(n)} + V_{u,1}^{(n)} - V_{u,4}^{(n)} + V_{diff} \quad (14)$$

$$V_{mft_{s,i}}^{(k)} = V_L - V_{u,6}^{(k)} \quad (15)$$

$$V_{mft_{s,o}}^{(k)} = V_L - V_{u,5}^{(k)} \quad (16)$$

The MFT electrical parameters (L_p , R_p , L_s , R_s , L_m) are determined through open circuit tests using a magnetostatic

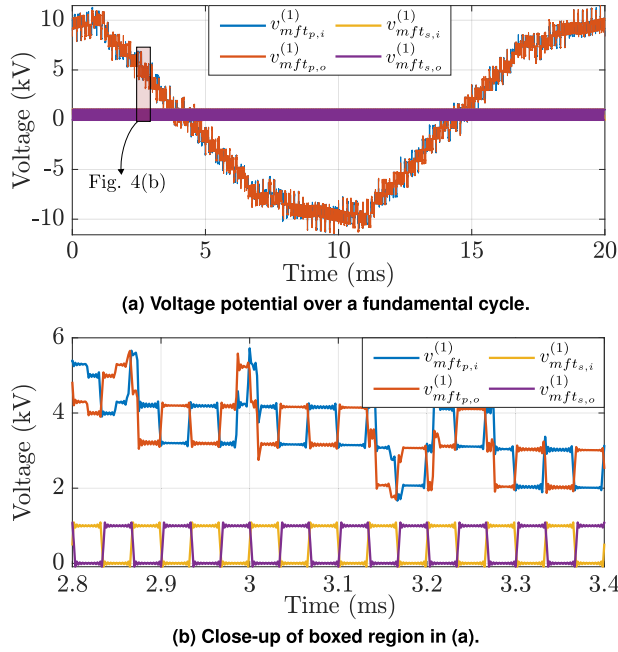


FIGURE 4. Module 1 MFT windings input and output voltage potentials.

TABLE 3. Reference MFT parameters.

Main MFT parameters			
Property	Symbol	Value	Unit
Rated voltage	$V_{r,MFT}$	1	kV
Primary leakage inductance	L_p	4.66	μH
Secondary leakage inductance	L_s	5.02	μH
Magnetizing inductance	L_m	2.2	mH
Turns ratio	N_t	1:1	-
Heatsink thermal resistance	-	0.1102	$^{\circ}\text{C/W}$
MFT core properties			
Property	Symbol	Value	Unit
Shape	-	U	-
Material	-	ferrote	-
Height	-	75	mm
Length	-	130	mm
Depth	-	30	mm
Effective area	A_e	840	mm^2
Number of cores in z-direction	N_{core}	13	-
MFT windings properties			
Property	Symbol	Value	Unit
Conductor type	-	litz wires	-
Strand material	-	copper	-
Strand diameter	d_{str}	0.2	mm
Number of Strands per wire	n_{str}	1664	-
Wire cross-section	w_{sec}	52.28	mm^2
Wire fill factor	ff	0.45	-
Wire diameter	$W_{d(p,s)}$	12.2	mm
MFT insulation properties			
Property	Symbol	Value	Unit
Wire insulation type	-	Polyimide (PI) film	-
Wire insulation thickness	-	0.2	mm
Wire insulation class	-	H	-
Winding insulation type	-	Epoxy resin	-
Winding insulation class	-	F	-
Interwinding insulation distance	t_{insw-w}	39.2	mm
Primary winding-core distance	t_{insw_p-c}	7.0	mm
Secondary winding-core distance	t_{insw_s-c}	2.7	mm
Air gap thickness	t_{gap}	0	mm

model with open boundary conditions, deployed in Finite Element Method Magnetics (FEMM) software [55]. The MFT waveforms ($v_{m,(p,s)}$, $i_{(p,s)}$) at rated power are then

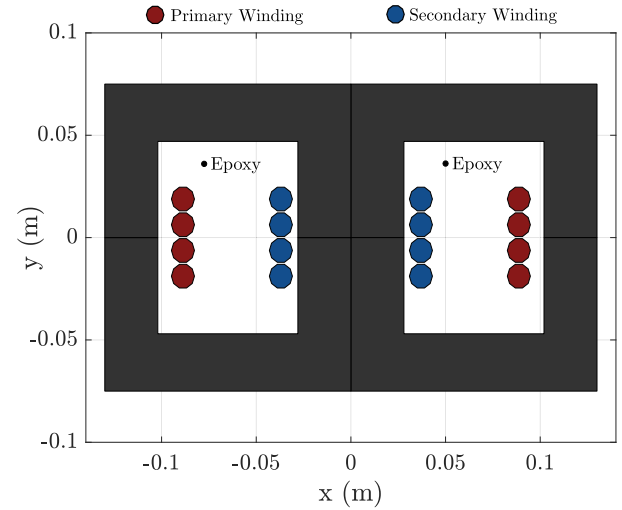


FIGURE 5. 2D cross-section of reference MFT design.

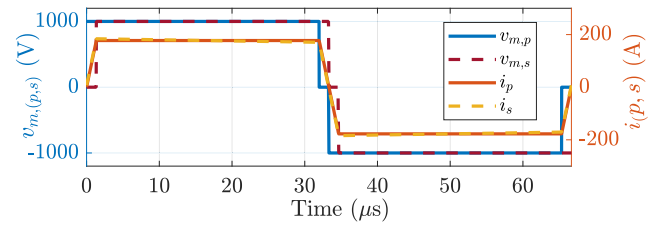


FIGURE 6. Modulated voltages and currents of reference MFT design at nominal power.

calculated, taking into account the specific isolated converter operation, to determine the MFT losses and the expected hotspot temperature. The criterion for a valid design in this work is set at 150°C at rated power. The hotspot estimation and power loss methodology are explained in more detail in the following section. In this study, the chosen transformer from the iterations is one characterized by relatively small leakage inductance and relatively high magnetizing inductance to achieve the highest efficiency using trapezoidal DAB modulation, as illustrated in [47]. In addition, the MFT design considers two aluminum heatsinks mounted on the top and bottom surface of the core area. The designed MFT parameters used in this study are presented in Table 3, and a 2D cross-section of the MFT is shown in Fig. 5. Additionally, the designed MFT waveforms at nominal power are presented in Fig. 6 and are utilized for the MFT losses calculation in the subsequent section.

IV. LOSS CALCULATION AND THERMAL ANALYSIS

Power electronic converters face two main stress factors throughout operation: electrical loading and temperature cycling [56]. These factors induce thermal stresses, necessitating careful design to ensure component durability. To evaluate the lifespan of each component under their experienced thermal conditions, precise temperature predictions are essential. In this study, owing to their low computational demands, electrothermal models are deployed

to estimate the hotspot temperatures of the SST components. In these models, the temperatures at the nodes are assumed homogeneous, with the temperature drops between them modeled via thermal resistances.

A. SEMICONDUCTORS LOSSES AND THERMAL MODELLING

Semiconductor losses are categorized into conduction and switching losses, calculated in this work as detailed in [47]. Fig. 7 illustrates the current division within a conventional power module configuration representing either a Si-IGBT or a SiC MOSFET. In the figure, i_m represents the modulated current, while i_{tr} and i_D denote current passing through each of the transistor (either an IGBT or MOSFET) and the diode, respectively. Whereas, the subscripts u and l refer to the upper and lower switches, respectively. The gating signal applied to the power module is represented by u_{sw} . Additionally, in this study, we consider zero dead-time for simplification.

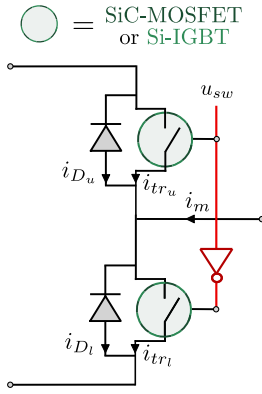


FIGURE 7. Single power module representation.

To determine conduction losses, 2-D linear scattered interpolators are employed to obtain the on-state voltages v_{on} and the forward voltages v_f for the transistors and diodes, respectively. These interpolation functions are established from the power module's data sheet and are a function of the respective component's current $i_{(tr,D)}$ and junction temperature $T_{j,(sw,D)}$. The conduction current is first calculated based on the switch technology. For an IGBT, the current is divided across each of the upper and lower transistors and diodes as follows.

$$\begin{aligned} i_{tr,u}^{igbt}(t) &= \begin{cases} -u_{sw}(t) \cdot i_m(t), & \text{if } i_m < 0 \\ 0, & \text{otherwise.} \end{cases} \\ i_{D,u}^{igbt}(t) &= \begin{cases} -u_{sw}(t) \cdot i_m(t), & \text{if } i_m > 0 \\ 0, & \text{otherwise.} \end{cases} \\ i_{tr,l}^{igbt}(t) &= \begin{cases} (1 - u_{sw}(t)) \cdot i_m(t), & \text{if } i_m > 0 \\ 0, & \text{otherwise.} \end{cases} \\ i_{D,l}^{igbt}(t) &= \begin{cases} (1 - u_{sw}(t)) \cdot i_m(t), & \text{if } i_m < 0 \\ 0, & \text{otherwise.} \end{cases} \end{aligned} \quad (17)$$

whereas for a SiC MOSFET power module, considering zero dead-time and owing to the third quadrant operation of the

transistor, the current flows in both positive and negative direction through the transistor. In which case, the current separation can be formulated as follows.

$$\begin{aligned} i_{tr,u}^{mosfet}(t) &= -u_{sw}(t) \cdot i_m(t) \\ i_{D,u}^{mosfet}(t) &= 0 \\ i_{tr,l}^{mosfet}(t) &= (1 - u_{sw}(t)) \cdot i_m(t) \\ i_{D,l}^{mosfet}(t) &= 0 \end{aligned} \quad (18)$$

The average conduction power losses for the upper and lower switches over a fundamental period T_0 can be computed as presented in (19), for both IGBTs and MOSFETs.

$$\begin{aligned} P_{tr(u,l)}(t) &= \frac{1}{T_0} \int_0^{T_0} i_{tr(u,l)}(t) v_{on}(i_{tr(u,l)}(t), T_{j,tr(u,l)}(t)) dt \\ P_{D(u,l)}(t) &= \frac{1}{T_0} \int_0^{T_0} i_{D(u,l)}(t) v_f(i_{D(u,l)}(t), T_{j,D(u,l)}(t)) dt \end{aligned} \quad (19)$$

For switching losses, the exact switching instances are determined as follows for each of the upper and lower Si-IGBTs.

$$\begin{aligned} t_{tr,u,ON}^{igbt} &= t_s, \text{ if } \Delta u_{sw}(t_s) > 0 \text{ \& } i_m(t_s) < 0 \\ t_{tr,u,OFF}^{igbt} &= t_s, \text{ if } \Delta u_{sw}(t_s) < 0 \text{ \& } i_m(t_s) < 0 \\ t_{D,u,OFF}^{igbt} &= t_s, \text{ if } \Delta u_{sw}(t_s) < 0 \text{ \& } i_m(t_s) > 0 \\ t_{tr,l,ON}^{igbt} &= t_s, \text{ if } \Delta u_{sw}(t_s) < 0 \text{ \& } i_m(t_s) > 0 \\ t_{tr,l,OFF}^{igbt} &= t_s, \text{ if } \Delta u_{sw}(t_s) > 0 \text{ \& } i_m(t_s) > 0 \\ t_{D,l,OFF}^{igbt} &= t_s, \text{ if } \Delta u_{sw}(t_s) > 0 \text{ \& } i_m(t_s) < 0 \end{aligned} \quad (20)$$

Here, t_s represents the specific time step of the simulation and Δu_{sw} is the difference in the gating signal from the previous time step. On the other hand, for SiC MOSFETs, as conduction occurs only through the transistor, switching losses are incurred only by the MOSFET at the following instances, while the body diode does not contribute to switching losses.

$$\begin{aligned} t_{tr,u,ON}^{mosfet} &= t_s, \text{ if } \Delta u_{sw}(t_s) > 0 \\ t_{tr,u,OFF}^{mosfet} &= t_s, \text{ if } \Delta u_{sw}(t_s) < 0 \\ t_{D,u,OFF}^{mosfet} &= 0 \\ t_{tr,l,ON}^{mosfet} &= t_s, \text{ if } \Delta u_{sw}(t_s) < 0 \\ t_{tr,l,OFF}^{mosfet} &= t_s, \text{ if } \Delta u_{sw}(t_s) > 0 \\ t_{D,l,OFF}^{mosfet} &= 0 \end{aligned} \quad (21)$$

The switching energies ($E_{tr,ON}$, $E_{tr,OFF}$, and $E_{D,OFF}$) are determined at the switching instances using 2-D linear scattered interpolators, which are functions of the commutated currents and the junction temperature. The switching power losses are then calculated from the interpolated energies as follows, where t_{step} denotes the simulation time step.

$$\begin{aligned} P_{tr(u,l),ON}(t_{tr(u,l),ON}) &= E_{tr(u,l),ON} \\ &\times (i_m(t_{tr(u,l),ON}), T_{j,tr(u,l),ON}) / t_{step} \end{aligned}$$

$$\begin{aligned}
P_{I_{r(u,l)OFF}}(t_{I_{r(u,l)OFF}}) &= E_{I_{r(u,l)OFF}} \\
&\times (i_m(t_{I_{r(u,l)OFF}}), T_{j,tr}(t_{I_{r(u,l)OFF}})) / t_{step} \\
P_{D(u,l)OFF}(t_{D(u,l)OFF}) &= E_{D(u,l)OFF} \\
&\times (i_D(t_{D(u,l)OFF}), T_{j,D}(t_{D(u,l)OFF})) / t_{step}
\end{aligned} \quad (22)$$

The electrothermal model of a power module is presented in Fig. 8, which parameters are extracted from the semiconductor's datasheet. The circuit network utilizes a Cauer model that reflects the physical structure of the semiconductor, thereby illustrating the internal temperatures of the layer sequence [57]. Since the loss calculations and the thermal model are interdependent, an initial junction temperature is assumed to calculate the losses and the resulting temperature from these losses. This temperature is then fed back into the loss calculation until a steady-state temperature is achieved. In this study, we consider three vacuum-brazed cold plates per SST module provided by [58], with the power modules of each FB-VSC mounted on a separate plate. Additionally, for simplicity, we assume a constant flow rate of 3.5 l/min, with a setpoint of 50°C.

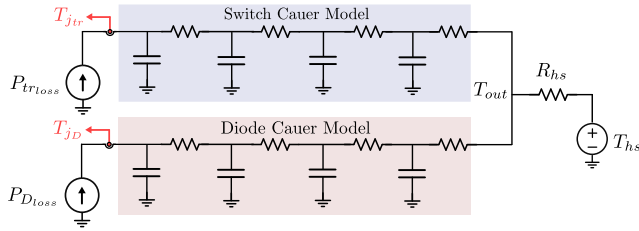


FIGURE 8. Power semiconductor electrothermal model.

B. CAPACITORS LOSSES AND THERMAL MODELLING

The power losses in capacitors can be calculated as the sum of the products of the capacitor's RMS current and its Equivalent Series Resistance (ESR) at each harmonic frequency.

$$P_{C_{loss}} = \sum_{k=1}^{N_h} ESR(k \omega_0) \cdot (|I_{cap}^{rms}(k \omega_0)|)^2 \quad (23)$$

In (23), N_h is the maximum considered harmonic order, ω_0 represents the fundamental angular frequency, whereas ESR is determined as follows

$$ESR(\omega) = R_s + \frac{\tan(\delta_0^C)}{\omega C} \quad (24)$$

where R_s is the sum of Ohmic resistances inside the capacitor, $\tan(\delta_0^C)$ signifies the dielectric dissipation factor, and ω is the operating angular frequency [51].

To estimate the hotspot temperature of the capacitor (T_C), the power losses are used in conjunction with a steady-state electrothermal model, expressed in (25), where the thermal

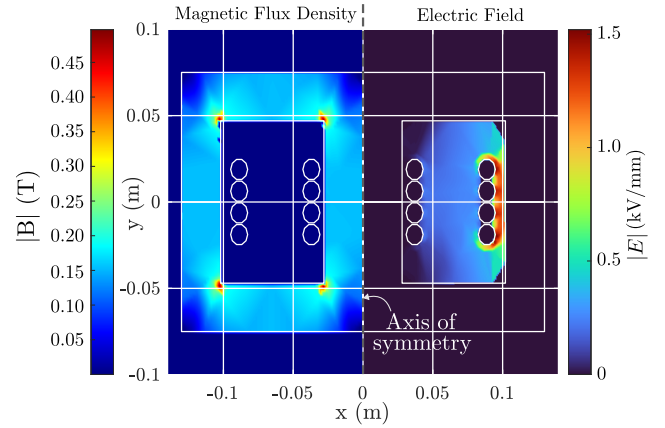


FIGURE 9. Magnetic flux density and electric field distribution with fundamental harmonic voltage excitation at P_{nom} .

resistance (R_{thC}) is extracted from the capacitor's datasheet.

$$T_C = T_{amb} + R_{thC} \cdot P_{C_{loss}} \quad (25)$$

C. MFT LOSSES AND THERMAL MODEL

MFT losses are classified into three main types: core losses, winding losses, and dielectric losses, with the core and winding losses making up the majority. In this section, the MFT of Module 1 is investigated at nominal power as a reference for the losses assessment.

Core losses are determined using Improved Generalized Steinmetz equation [59], given by

$$P_c = \sum_{s=1}^{N_s} \frac{1}{T} \int_0^T k k_i \left| \frac{dB_s(t)}{dt} \right|^\alpha \Delta B_s^{\beta-\alpha} dt \quad (26)$$

where,

$$k_i = \frac{1}{(2\pi)^{\alpha-1} \int_0^{2\pi} |\cos(\theta)|^\alpha 2^{\beta-\alpha} d\theta} \quad (27)$$

The Steinmetz parameters k , α , and β are used along with the instantaneous and peak-to-peak magnetic flux density, $B_s(t)$ and ΔB_s , for each finite element s out of a total N_s . A magnetostatic model with open boundaries, implemented in FEMM, calculates the magnetic flux density amplitude $\hat{B}_s(h\omega_0)$ for each harmonic h of the fundamental frequency ω_0 . This harmonic data reconstructs the time-domain waveform used in (26). Fig. 9 presents the maximum absolute value of the magnetic flux density over one fundamental period for the waveforms shown in Fig. 6.

The winding losses are computed by summing the losses for each individual harmonic as follows

$$P_{w(p,s)} = \sum_{h=1}^{N_h} R_{dc,w(p,s)} F_r(h \omega_0) I_{w(p,s)}^{(h)2} \quad (28)$$

where $R_{w(p,s)}$ represents the DC resistance of the winding, F_r is the resistance factor for round litz wires from the modified Dowell's equation, linking DC to AC resistance at

each harmonic [59], while $I_{w(p,s)}^{(h)}$ denotes the RMS amplitude of single-sided current harmonics.

Dielectric losses depend on the frequency, the RMS amplitude of the single-sided electric field harmonics ($E_{ins}^{(h)}$), the volume of the dielectric region (Vol), and the material properties that vary with frequency, specifically absolute electric permittivity (ϵ_{ins}) and the loss coefficient ($\tan(\delta_{ins}^{(h)})$). This relationship is illustrated in (29) [60].

$$P_{ins} = \sum_{h=1}^{N_h} \iiint_{Vol} \epsilon_{ins} \cdot \tan(\delta_{ins}^{(h)}) \cdot h \omega_0 \cdot E_{ins}^{(h)2} dVol \quad (29)$$

In this work, the dielectric losses are assessed using a current flow model with open boundary conditions in FEMM, with the core at ground potential (0 V). The model is excited with fixed voltage conditions at each conductor contour. The resulting electric field distribution, as depicted in Fig. 9 for the reference MFT at the fundamental harmonic excitation (f_0 component of the waveforms presented in Fig. 4), along with the current density are extracted from the FEMM model and are used to calculate the power loss density in the insulating material, representing the dielectric losses.

The steady-state thermal model utilized in this study is based on the adaptive MFT thermal model presented in [61], which is devised including conduction, convection, and radiation mechanisms. This thermal model, depicted in Fig. 10, was bench-marked against a heat run test applied to a DAB converter test bench in [61], operating with conventional rectangular modulation with the parameters described in Table 4. The schematic of the tested DAB converter, along with the experimental setup and the associated MFT, are presented in Fig. 11. This test serves to validate the MFT losses calculations which are translated into heat injections to their corresponding nodes emulated by current sources. The results of the heat-run showed a maximum error around 16% at all tested operating conditions [61].

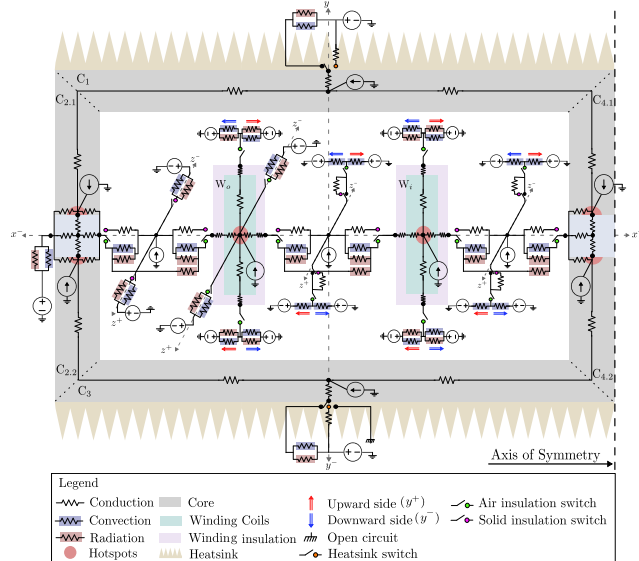


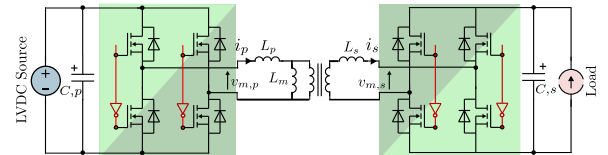
FIGURE 10. Steady state thermal network of shell-type MFT.

TABLE 4. DAB test bench parameters.

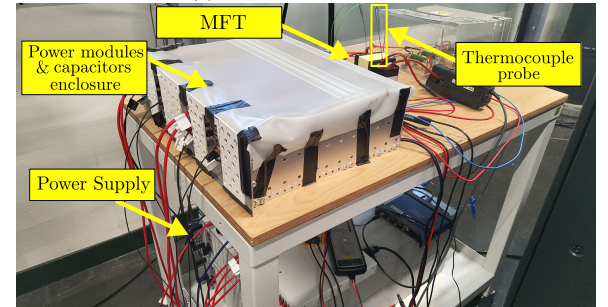
Parameters	Value	Units
Nominal power (P_{nom})	5	kW
Nominal primary bus voltage ($V_{dc,p}$)	400	V
Nominal secondary bus voltage ($V_{dc,s}$)	393	V
Primary bus capacitance ($C_{p,p}$)	1	mF
Secondary bus capacitance ($C_{s,s}$)	1	mF
Switching Frequency (f_{sw})	20	kHz
MFT primary leakage inductance (L_p)	5.80	μ H
MFT secondary leakage inductance (L_s)	5.64	μ H
MFT magnetizing inductance (L_m)	4.2	mH
MFT turns ratio (N_r)	1	-

V. COMPONENTS LIFETIME EVALUATION

To assess the impact of different stress cycles on each component within the SST, accelerated life tests should ideally be performed on multiple samples of these components to estimate the time or cycles to failure under said stress. However, due to minor manufacturing variations, accurately predicting the precise failure time of an individual device is impractical. Therefore, it is more meaningful to determine the failure probability of each device. Consequently, for the wear-out failure in this study we employ Monte Carlo analysis to account for uncertainties in both the lifetime equation parameters as well as the system measurements. This methodology is detailed in the reliability estimation studies described in [36], [42], and [62]. The total damage experienced by the device due to various stresses throughout



(a) DAB circuit schematic.



(b) DAB test bench.



(c) Investigated MFT for the thermal circuit verification.

FIGURE 11. MFT heat run test experimental setup.

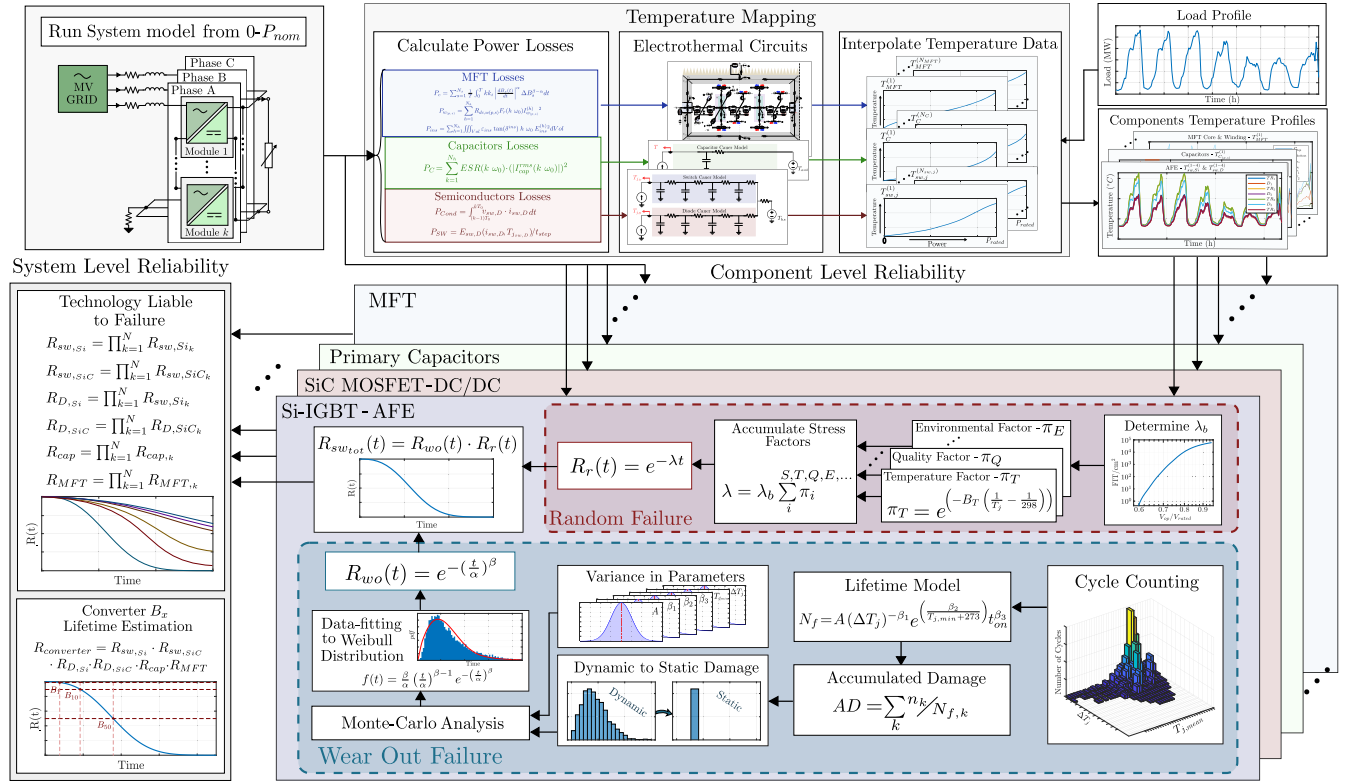


FIGURE 12. SST mission profile based lifetime estimation algorithm.

the load profile is first calculated. These different stresses are then converted into an equivalent single stress that would produce the same calculated total damage. Next, uncertainties in the lifetime equation parameters are taken into account, considering the equivalent single stress values, through the Monte Carlo analysis. The results of this analysis are subsequently used to fit the *pdf* of the Weibull function presented in (7). While a case study of a PV inverter in [63] demonstrated that Monte Carlo reliability assessment results generally converge when the number of simulations reaches 1000, this work sets the sample size to 10000. This number, initially introduced in a degradation assessment of IGBTs by [7], is considered sufficiently large to ensure convergence regardless of the parameter variation range. The simulations in this study were conducted on a Windows 10 Pro system with an 11th Gen Intel(R) Core(TM) i7-11850H @ 2.50 GHz processor, with each Monte Carlo run of the specified sample size taking no more than 3 seconds.

The complete procedure for calculating the mission profile based SST reliability, as illustrated in Fig. 12, involves evaluating system behavior at various operating points up to P_{rated} and determining the losses and temperatures of each device at these corresponding points. This “temperature mapping” method simplifies the analysis by directly linking the thermal response of components to a specific mission profile. While seasonal ambient temperature variations and different AC input voltage distortions (such as voltage unbalance, harmonics, and DC offsets, etc.) can be considered

at this stage, this study assumes a constant input voltage to the SST and a steady ambient temperature of 27°C for simplicity.

Once the temperature profile for each component is established, the subsequent steps for calculating component reliability are detailed in the following subsections. The reliability of the SST depends on each individual component; any failure (eg. switch, capacitor, or MFT) can lead to complete converter failure in the absence of redundant modules. Therefore, the total SST reliability, accounting for both wear-out and random failures, is expressed as the product of each component’s reliability, as illustrated in Fig. 12.

A. SEMICONDUCTORS RELIABILITY ESTIMATION

The number of cycles to failure (N_f) of semiconductors are influenced by the specific lifetime model applied. Whereas the lifetime model and its parameters are specific to the power modules packaging technology. While an ideal model would reflect the actual testing conditions encountered in the field, this is often impractical due to the significant time required for such testing. Therefore, as this study employs the standard multi-layer structure power modules, we utilize the widely accepted CIPS model [8], detailed in (30), which is devised through accelerated lifetime testing of multiple IGBTs, constituting seven different bonding technologies and seven different packaging. This empirical equation addresses the key failure mechanisms in IGBTs: bond-wire lift-off and solder joint fatigue [7], [42]. A 5% to 20% increase in forward

voltage drop typically signals bond-wire lift-off, whereas a 20% to 50% rise in thermal resistance between the chip and heatsink indicates solder fatigue. The latter is characterized by crack propagation within the solder layers [64]. Short temperature cycles (t_{on}) occurring over just seconds or less mainly contribute to bond wire degradation [65], [66]. In contrast, longer t_{on} contribute more to the base plate solder degradation. Thus, it is necessary to include the effects of the temperature cycle length on the number of cycles to failure which is modelled as presented in (31) [65], [66]. As the sampled load data in this study contribute to long temperature cycles, solder joint fatigue are considered the more prominent failure mechanism.

$$N_f = A \cdot (\Delta T_j)^{-\beta_1} \exp\left(\frac{\beta_2}{T_{j,min} + 273}\right) \cdot t_{on}^{\beta_3} \quad (30)$$

$$\frac{N_f(t_{on})}{N_f(1.5)} = \begin{cases} 2.25, & t_{on} \leq 0.1 \text{ s} \\ \left(\frac{t_{on}}{1.5}\right)^{-0.3}, & 0.1 \text{ s} \leq t_{on} \leq 60 \text{ s} \\ 0.33, & t_{on} \geq 60 \text{ s} \end{cases} \quad (31)$$

In (30), ΔT_j is the fluctuation in junction temperature, $T_{j,min}$ is the minimum junction temperature, whereas A , β_1 , β_2 , and β_3 are the lifetime equation coefficients [8].

Although initially developed for Si-IGBTs, the CIPS model has also been evaluated for SiC-MOSFETs [9], [10], given the similarities in their failure modes. In SiC-MOSFETs, the chip die is soldered to the copper base, with common failure location in standard packaging involving wire bonding solder and die attach. These components experience degradation as a result of temperature cycling and mechanical stress [11], [12], [67]. The CIPS model in this paper is selected for SiC MOSFETs based on the selected power modules and their dominant failure mechanisms. Notably, with the development of advanced packaging technologies, such as sintered chip modules like the SKiM [68], traditional soldering techniques have been replaced by newer methods, such as silver diffusion bonding. These modules failure modes primarily exhibit bond wire lift-off and heel cracking. For such cases, the Semikron lifetime model proposed in [69] offers a more accurate degradation assessment.

To analyze the Accumulated Damage (AD) of the semiconductors, this paper applies Miner's rule based on the assumption of linear cumulative damage and the independence of semiconductor's damage from the order of the stress cycles [70]. The AD is calculated as the sum of the ratios of the number of cycles experienced by the semiconductor under each specific stress type (n_k) to the corresponding number of cycles to failure for that stress type ($N_{f,k}$), where k denotes different stress cycle types.

$$AD = \sum_k \frac{n_k}{N_{f,k}} \quad (32)$$

The uncertainties in the semiconductors lifetime model include, the lifetime equation parameters A , β_1 , β_2 , and β_3 ,

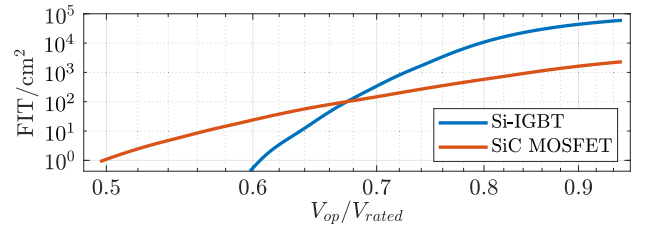


FIGURE 13. Averaged FIT rates of semiconductor switches normalized to chip area [71], [72], [73], [74], [75].

as well as variations in each of $T_{j,min}$ and ΔT_j resulting from deviations of the on-state voltage of the semiconductor calculated similarly as presented in [7].

To calculate the random failure rate of the semiconductors, (6) is employed. However, base failure rates, also known as Failure in Time (FIT) rates, are often not readily available in the datasheets. Therefore, an average of the failure rates reported in the literature and by different semiconductor suppliers across various rated voltages was calculated [71], [72], [73], [74], [75]. These average FIT rates are presented in Fig. 13, serving as the base failure rate which already accounts for the effects of electric stress. Additionally, to incorporate the impact of thermal stress on the random failure rate, the following equation is utilized, sourced from MIL-HDBK 217 [41], where, B_T represents a constant that corresponds to the activation energy divided by the Boltzmann constant.

$$\pi_T = \exp\left(-B_T \left(\frac{1}{T_j} - \frac{1}{298}\right)\right) \quad (33)$$

B. CAPACITOR RELIABILITY ESTIMATION

The wear out failure mode of film capacitors is determined when one of its electric parameters, namely Capacitance (C), Equivalent Series Resistance (ESR), dielectric loss factor ($\tan(\delta_0^C)$), leakage current, or insulation resistance, drifts over a specific threshold from its initial value [4]. These drifts are mainly triggered through breakdowns of the dielectric material from continuous electrical and thermal stresses. The state-of-the-art lifetime equation for film capacitors accounting for these failure mechanisms and relating lifetime in hours to applied voltage and hotspot temperature is given by

$$L = L_0 \left(\frac{V}{V_0}\right)^{-n_1} 2^{\frac{T_0 - T}{n_2}} \quad (34)$$

where n_1 and n_2 are the voltage stress exponent and the temperature stress constant, respectively. L , V , and T are the lifetime, voltage, and temperature in Kelvin under the use condition, respectively; whereas, L_0 , V_0 , and T_0 are the lifetime, voltage, and temperature in Kelvin at the test condition, respectively. Following the “10-Kelvin-rule” of the Arrhenius law, n_2 equals 10 for the analyzed capacitors [62]. Typically, for film capacitors n_1 is set to 9 [4], [5], however, it is essential to note that n_1 is dependent on the

voltage stress level [4], [6]. In case of low voltage stress, n_1 is adjusted to 1, such that the voltage stress is directly expressed as the ratio of use voltage to tested voltage.

The consumed lifetime of a capacitor can be calculated using

$$CL = \sum_{i=1}^N \frac{t_i}{L_i} = \sum_{i=1}^N \frac{t_i}{L_0 \left(\frac{V_i}{V_0} \right)^{-n_1} 2^{\frac{T_0 - T_i}{n_2}}} \quad (35)$$

where t_i is the duration of operation with temperature T_i and voltage V_i , at stress point i . The lifetime of the capacitor can be calculated by the reciprocal of CL . The failure distribution function is obtained in a similar manner to the semiconductors by deploying Monte Carlo analysis. Following the approach presented in [5], two types of uncertainties are considered: 1) lifetime model-related parameter, n_1 and n_2 , and 2) variations due to manufacturing processes, L_0 and T_0 .

For random failures, the base failure rates (λ_b) of the capacitors and the corresponding effects of the hotspot temperature and the operating voltage are obtained directly from the capacitors' respective datasheets. The random failure rate is calculated as the average of the various FIT rates corresponding to each load point.

C. MFT RELIABILITY ESTIMATION

Specific failure data on MFTs is limited, however, research on distribution transformers can be used to highlight similar stress factors. Common points of failure in distribution transformers include bushings, internal connections, and inter-turn and winding insulation breakdown [76]. The structural similarities between distribution transformers and MFTs, coupled with the adverse effects of high frequencies on insulation degradation [32], [33], [34], [35], suggest that insulation failures are a predominant concern for MFTs. As insulation is subjected to thermal, electrical, and mechanical stresses, failures may arise from material degradation over time or through faults such as overvoltage or short circuits [76]. High-frequency stresses can exacerbate dielectric heating and Partial Discharge (PD), increasing the risk of insulation breakdown [32], [33], [60]. Therefore, a thorough examination of these failure mechanisms is essential for accurately estimating the lifespan of MFTs.

The MFT reliability model in this study is based on the Multiconductor Transmission Line (MTL) principles presented in [33]. MTL models discretize electromagnetic equations of a coupled network that captures mutual conductor interactions. Within MFTs, MTL capacitors represent energy stored in insulations, influencing performance through high-frequency losses, PD, and insulation issues. An example of the MTL parasitic capacitance model for a 3:3 turn ratio MFT is shown in Fig. 14.

Capacitor and insulation lifetime models share similar dependencies on electrical and thermal stresses, primarily due to dielectric degradation, which leads to punctures and increased losses [4], [77]. Insulation failure is often

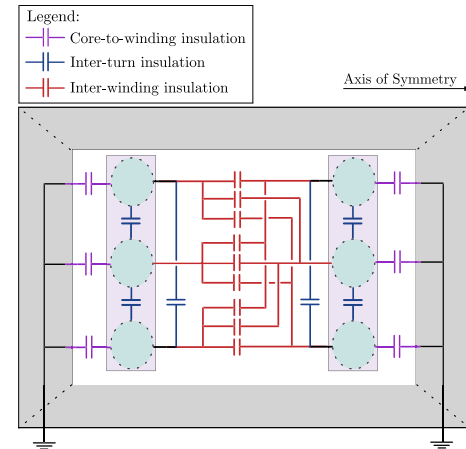


FIGURE 14. MFT insulation division into multiple dielectric regions (adapted from [33]).

associated with PD issues, analyzed through exponential or inverse power law models [78]. In solid-insulated transformers, PD is amplified by voids in the insulation that can provide discharge pathways, increasing failure risk. In addition, variations in the thermal expansion coefficients of adjacent materials can cause cracks or delamination, further elevating PD activity [79]. Thus, the characteristics of the voids within the insulation significantly affect PD activity, as it determines the Partial Discharge Inception Voltage (PDIV), the threshold for PD initiation. Frequent PD events reduce insulation longevity, with the PDIV marking a critical stress threshold that triggers electrical aging.

The reliability model for MFTs is designed as a series network that combines the reliabilities of all the dielectric regions, illustrated as parasitic capacitors in the MTL model. The lifetime model for each region, presented in (36), was devised for insulating materials in [80] and further validated in [81] and forms the basis of the capacitor lifetime model, which is simplified in (34) [4]. In (36), E_a represents the activation energy while K_b is the Boltzmann constant. The consumed lifetime for each dielectric region can be calculated similarly to the approach used in (35).

$$L = L_0 \left(\frac{V}{V_0} \right)^{-n} \exp \left[\left(\frac{E_a}{K_b} \right) \left(\frac{1}{T} - \frac{1}{T_0} \right) \right] \quad (36)$$

Thermal stresses on the MFT insulating material are derived from the nodes of the electrothermal model representing the stressed regions. Whereas regarding the electric stress, the voltage potentials of each winding turn are first calculated with reference to ground. In an ISOP configuration of modules, the first module of each phase experiences the highest voltage stresses. As one moves down the module, each subsequent module has its input voltage potential with respect to ground reduced by $v_m^{(k)}$. Regarding the MFT insulation, the MV side winding terminals ($v_{mftp,i}^{(k)}$ and $v_{mftp,o}^{(k)}$, as denoted in Fig. 2 are determined by the difference between the module's AC input voltage referenced to ground and the voltage drops across the switches along the path to the

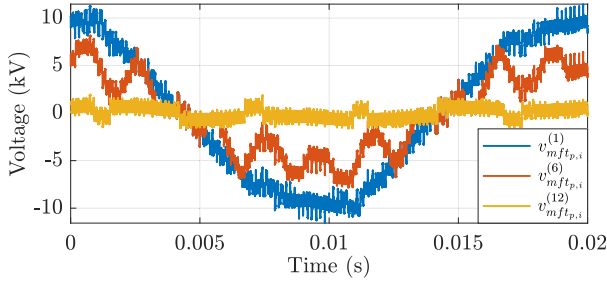


FIGURE 15. Time-domain waveforms of MFT primary winding input voltage potentials to ground for modules 1, 6, and 12.

MFT as described in 13 and 14. Fig. 15 shows the time domain waveforms of $v_{mft_{p,i}}^{(k)}$ for modules 1, 6, and 12 of a single phase for the studied SST. The voltage harmonics of the first module display the 50 Hz component of the grid voltage along with high-frequency harmonics associated with the converter's switching frequency. Moving down the modules of the phase, the dominance of the first-order harmonic (i.e., 50 Hz) diminishes, while the amplitudes of the converter's switching frequency harmonics become relatively more pronounced. Hence, in order to facilitate the manufacturing and maintenance of the MFT, the MFTs in ISOP-configured SSTs are typically designed based on the module experiencing highest insulation stress (module 1). With known terminal voltages, individual wire turn voltages are estimated assuming a uniform voltage gradient across the adjacent conductors as described in Section III-C.

In (36), V_0 represents the electric stress threshold voltage, indicating the PDIV of the insulating material. The PDIV estimation utilizes the modified Paschen's law, presented in (37), which incorporates a temperature correction factor [82]. This law describes the breakdown voltage ($V_{b,v}$) as a function of temperature (T) for specific gas and electrode materials under uniform electric field conditions, and it is dependent on gas property-specific parameters (A , B), gas pressure (P), and void length (d_v), as follows

$$V_{b,v} = \frac{B \cdot P \cdot \left(\frac{273+25}{273+T} \right)}{\ln \left(\frac{A \cdot P \cdot d_v \cdot \left(\frac{273+25}{273+T} \right)}{k} \right)} \cdot d_v \quad (37)$$

where k is a constant calculated as

$$k = \ln \left(1 + \frac{1}{\gamma} \right) \quad (38)$$

The coefficient γ represents Townsend's secondary emission coefficient, indicating the average number of electrons released from the cathode per incident positive ion [82], [83], [84]. The temperature variation of γ for the transformer's insulating materials is sourced from [82] for epoxy and [83] for polyimide (PI). In addition, this study assumes the gas in the voids is air at a standard atmospheric pressure ($P=101,325$ Pa), with $A = 11.25$ Pa m^{-1} and $B = 273.75$ V Pa $^{-1}$ m^{-1} [84].

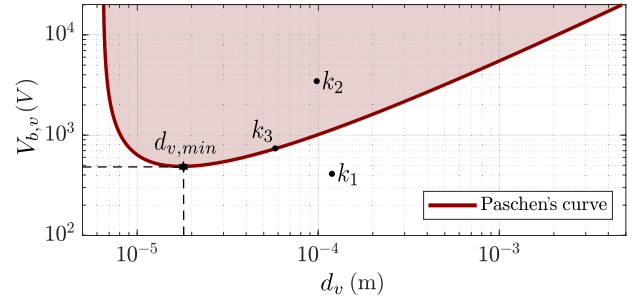


FIGURE 16. Paschen's curve at ambient pressure and temperature ($V_{b,v}$ is peak voltage).

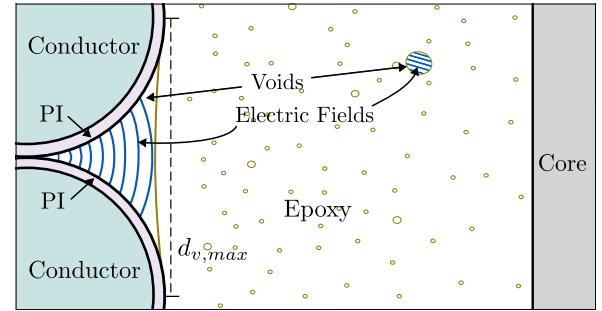


FIGURE 17. Illustration of the possible voids composition inside solid-insulated MFT (not to scale).

Fig. 16 shows Paschen's curve for an air gap at atmospheric pressure and room temperature, indicating three conditions: 1) No PD activity below the curve (e.g., k_1), 2) PD activity within the shaded region (e.g., k_2), and 3) Points on the curve (e.g., k_3) representing the PDIV. Breakdown voltage sharply increases at relatively very small void lengths (d_v) due to insufficient gas molecules, while it rises linearly at higher d_v due to weaker electric fields [85].

Identifying potential air gap locations within the solid-insulated MFT is crucial; these can form both within the insulating material and between adjacent wires, as shown in Fig. 17. As high temperature is usually correlated to lower PDIVs, we can use the Paschen's curve at the highest expected temperature to estimate the minimal gap ($d_{v,min}$) likely to initiate PD activity, as presented in Fig. 16. On the other hand, following the approach presented in [86], the maximum length of a possible air gap ($d_{v,max}$) in this work is defined as twice the wire radius, applicable to both void types.

Once the range of possible void lengths is identified, and knowing the insulating layer length through which the electric

TABLE 5. Voltage exponential term gathered from the literature.

Material	Waveform Shape	n	Reference
PI	Sinusoid	2.15	[87]
Nomex Paper	Square	2.02	[88]
Epoxy	Pulse	2.64	[89]
Epoxy	Sinusoid	3.12 / 3.337	[90]
Epoxy	Square	2.06	[91]

field lines pass, we apply the dielectric boundary condition to calculate the maximum voltage across the voids of different lengths as follows

$$V_{void} = V_{op} \cdot \left(\frac{d_v}{\varepsilon_{r,v}} \right) / \left(\frac{d_v}{\varepsilon_{r,v}} + \frac{d_{ins}}{\varepsilon_{r,ins}} \right), \quad \{d_v \in \mathbb{R} \mid d_{v,min} \leq d_v \leq d_{v,max}\} \quad (39)$$

where d_{ins} is the length of the electric field path between the active electrodes, $\varepsilon_{r,v}$ and $\varepsilon_{r,ins}$ are the permittivities of the inner gas and the surrounding insulation, respectively, and V_{op} denotes the maximum operating voltage applied by the electrodes. By comparing the calculated voltages of d_v to Paschen's curve, we can assess the potential for PD activity. If the voltages do not intersect the Paschen curve, it indicates no electric stress under normal operation. Conversely, an intersection signifies that PD activity is possible, in which case the d_v representing the highest ratio of $V_{void}/PDIV$ is considered, as it represents the worst-case scenario corresponding to the lowest PDIV. When the voltage stress is below the PDIV, the exponential component n in (36) is set to zero to neglect the electric stress effects. If the voltage exceeds the PDIV, n is assigned a value of 2.55, representing the average from Table 5 which presents voltage exponent stress factors based on voltage-lifetime data from existing literature. A higher n reflects a greater impact of electric stress on insulation degradation. However, insulation in MFTs is typically designed with generous safety margins, ensuring operational voltages stay well below the material's breakdown strength. This justifies using a low n , consistent with standard capacitor lifetime models under low electric stress [4].

To estimate the MFT wear-out *pdf*, two types of uncertainties are addressed in the Monte Carlo analysis. The first involves parameters related to the lifetime model: n varies by 22%, with its lower bound matching the minimum value from Table 5, while E_a and L_0 both vary by 5%. The second are uncertainties in estimated values, including the temperature which varies by 16% reflecting the maximum error margin of the electrothermal model, Townsend's secondary emission coefficient (γ) varying by 5%, and the void length (d_v) which varies between the predefined minimum and maximum values.

For the MFT random failure computation, the base failure rate (λ_b) is typically sourced from the manufacturer's datasheet or approximated from MIL-HDBK-217 handbook [41]. The primary stress factor, as outlined in MIL-HDBK-217, is the hotspot temperature (T_{hs}) of the MFT.

Hence, the random failure rate (λ_r) which represents the mean of the instantaneous failure rate calculated at each operating point i across n total steps in the simulation, is determined using (40) for the studied insulation class [41].

$$\bar{\lambda}_r = \frac{1}{N} \sum_{i=1}^N \lambda_b \exp \left(\frac{T_{hsi} + 273}{409} \right)^{10} \quad (40)$$

VI. SST CONSUMED LIFETIME: CASE STUDIES

The distribution of losses within the system is critical to understanding its overall performance and reliability. The overall theoretical efficiency curve of the designed SST, considering the investigated system parameters and components as described in Section IV, is presented in Fig. 18 and the combined losses of each conversion stage per module are presented in Table 6. It is important to note that the losses and efficiencies reported are derived from suppliers data and have not been validated through laboratory testing. Additionally, the losses do not account for cooling equipment, power supplies, or other auxiliary components required for the SST operation. Furthermore, it is assumed that every instance of Zero Voltage Switching (ZVS) and Zero Current Switching (ZCS) of the trapezoidal operation of the DAB converter incurs zero power losses. However, this assumption is a simplification, as it is common to observe low (but non-zero) turn-on losses that increase at higher frequencies during ZVS operation. Whereas ZCS losses depend on the transformer magnetizing current, which may also result in hard-switching conditions.

The power loss results indicate that the efficiency drop is primarily due to the AFEC, followed by the DAB converter, the MFT, and finally the capacitors. The zero power losses of the SiC diodes in the DAB converter are attributed to

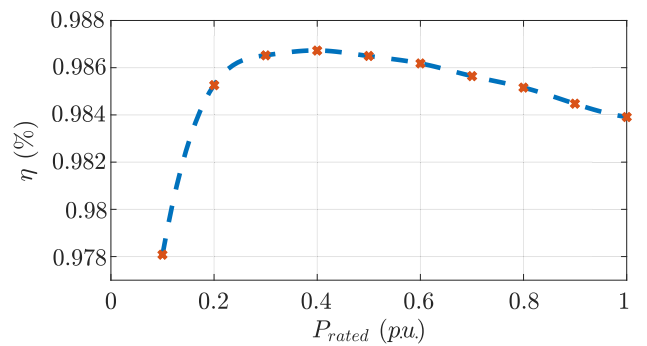


FIGURE 18. Designed SST efficiency (η) vs. power.

TABLE 6. SST modules losses distribution at rated power.

Stage	AFEC		DAB Conv. Prim. Bridge		DAB Conv. Sec. Bridge		Capacitors		MFT - Module 1				Total
Component	Transistor	Diode	Transistor	Diode	Transistor	Diode	Prim.	Sec.	Core	Prim. Winding	Sec. Winding	Dielectric	
Losses (W)	639.12	524.33	448.34	0	455.61	0	34.27	0.035	249.67	159.76	164.28	3.56	2679.0
MFT - Module 12													
									Core	Prim. Winding	Sec. Winding	Dielectric	
									249.67	159.76	164.28	1.28	2676.7

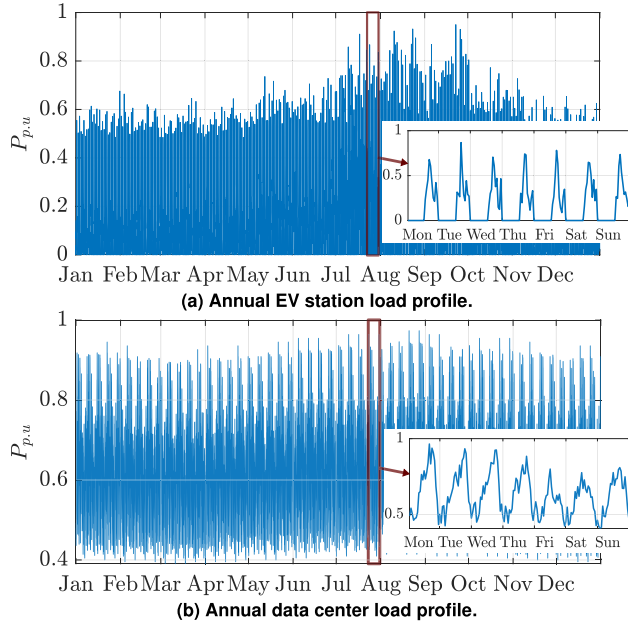


FIGURE 19. Investigated load profiles sampled hourly.

reverse current flow through the MOSFET channel, along with the assumption of lossless ZVS and ZCS conditions. This is justified as no dead-time is considered for the sake of simplicity. While the losses of each converter remain consistent across the modules, the effects of dielectric losses on the MFT vary due to the different voltage potentials across each module's insulation, as previously discussed. Notably, the difference in MFT losses between Modules 1 and 12 is ≈ 2.3 W, which is insignificant compared to the combined core and winding losses, which are orders of magnitude higher for the presented MFT design. Nevertheless, these losses contribute to direct heat generation in the insulating materials, potentially influencing aging. Therefore, this section investigates the impact of these losses on the reliability of the MFT, alongside a comprehensive reliability analysis of the SST, both with and without considering the MFT.

Two test cases, each associated with a distinct mission profile, are used to assess the SST lifetime. The first load profile in Fig. 19 represents an Electric Vehicle (EV) station charging load sourced from [92], while the second profile depicts a data center load reported in [93], both of which are sampled hourly. To accommodate load fluctuations throughout the year, seasonal variations are applied as well as up to $\pm 20\%$ of the maximum load is randomly assigned at different instances. The choice of these two loads is based on their differing annual and daily peak loads as well as their load durations. Moreover, reliability is significantly prioritized for data centers in comparison to EV charging loads due to the common high availability requirements, therefore assessing different reliability demands. It's important to highlight that the sampling rate is low enough to ignore the transient behavior between steady-state points of the MFT's temperature, thereby confirming the appropriateness of the electrothermal model for the load profile being analyzed.

TABLE 7. Lifetime models parameters and variations [4], [5], [7], [8], [9], [35], [51], [94], [95].

Si-IGBT Parameters			
Parameter	Value	Unit	Variance
A	$9.34e14$	-	-
β_1	-4.416	-	6.36 %
β_2	$1.285e3$	-	20.95 %
β_3	-0.463	-	23 %
$T_{j,min}$	Load profile dependent	$^{\circ}\text{C}$	Based on V_{CE} variation
T_j	Load profile dependent	$^{\circ}\text{C}$	Based on V_{CE} variation
SiC MOSFET Parameters			
Parameter	Value	Unit	Variance
A	$3.21e8$	-	-
β_1	-3.775	-	5 %
β_2	$2.750e3$	-	5 %
β_3	0	-	-
$T_{j,min}$	Load profile dependent	$^{\circ}\text{C}$	Based on V_{DS} variation
T_j	Load profile dependent	$^{\circ}\text{C}$	Based on V_{DS} variation
Capacitors Parameters			
Parameter	Value	Unit	Variance
L_0	$100e3$	hours	5 %
n_1	9	-	5 %
n_2	10	-	5 %
T_0	75	$^{\circ}\text{C}$	-
V_0	700	V	-
T_{mean}	Load profile dependent	$^{\circ}\text{C}$	5 %
MFT Parameters: Inter-turn Insulation (PI)			
Parameter	Value	Unit	Variance
L_0	$300e3$	hours	5 %
E_a	0.9278	eV	5 %
n	2.55	eV	22 %
T_{mean}	Load profile dependent	$^{\circ}\text{C}$	16 %
γ	Temperature dependent	-	10 %
d_v	-	m	$[d_{v,min}, d_{v,max}]$
MFT Parameters: Winding-core and Inter-winding Insulation (Epoxy)			
Parameter	Value	Unit	Variance
L_0	$330e3$	hours	5 %
E_a	0.77	eV	5 %
n	2.55	eV	22 %
T_{mean}	Load profile dependent	$^{\circ}\text{C}$	16 %
γ	Temperature dependent	-	10 %
d_v	-	m	$[d_{v,min}, d_{v,max}]$

For the components wear-out investigation, the parameters associated with the lifetime equations and the variations applied in the Monte Carlo analysis are detailed in Table 7. All parameters subjected to variation are represented using normal distribution functions with a 95% confidence interval, with the exception of d_v . As the average expected value for d_v is not specified in this research, its variation is instead modeled using a uniform distribution that spans from $d_{v,min}$ to $d_{v,max}$. Additionally, the deviations in the MFTs' V_0 parameter representing the PDIV are taken into account through the variations of d_v and γ . Consequently, a different PDIV is estimated for each sample of dielectric region within each MFT in the Monte Carlo analysis.

The results of the Monte Carlo analysis yield the failure probability density function $f(t)$ for each component, from which the cumulative probability of failure over time, $F(t)$, is derived. Fig. 20 illustrates the aggregated reliability $(1 - F(t))$ of the various component technologies within Module 1, analyzed under the two load profiles. AFE_{tr} represents the combined reliability of all the Si-transistors in the AFEC, while AFE_D denotes the combined reliability of the Si-diodes. Similarly, DC_p and DC_s correspond to the

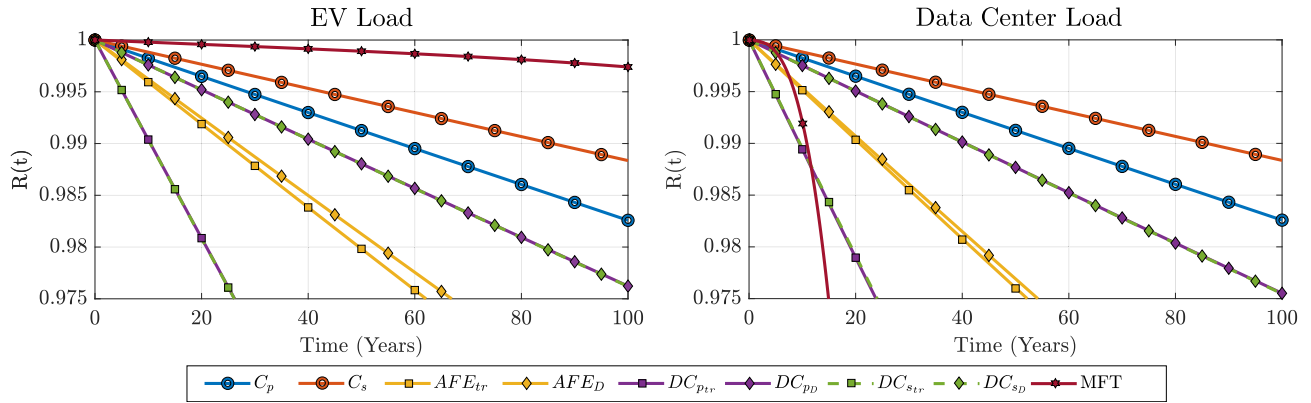


FIGURE 20. Module 1 components Reliability for each mission profiles.

primary and secondary side FB-VSCs of the isolated DC-DC converter, respectively, utilizing SiC technologies. Whereas C_p and C_s indicate the combined reliabilities of the capacitors forming the primary and secondary DC links, respectively.

It is anticipated that the devices within the data center module will experience greater wear than those under the EV load due to the higher continuous power demand. The identified high power loss components in Table 6, particularly the AFEC transistors and diodes, the SiC transistors of the DAB converter, and the MFT, indicate that they operate under increased stress on average when subjected to the data center load compared to the EV load. This is reflected in the reliability curves in Fig. 20 as their lifetime is relatively more affected by the change in load demand compared to the other components. However, the analysis shows that compared to the MFT, other devices exhibit only minor reliability variations due to safety margins in component selection. The MFT's unreliability relatively increases as a result of constant high-temperature operation exhibited with the data center load. With constant SST input voltage in the two scenarios, thermal stress is the main factor impacting the designed MFT reliability, underscoring the mission profile's influence on its lifetime.

Furthermore, to evaluate the sensitivity of the lifetime model to thermal conditions, a $\pm 5^\circ\text{C}$ variation in the cooling fluid set-point of the AFEC cooling plate is assessed considering the EV load. The results of this sensitivity assessment are provided in Fig. 21. The results indicate

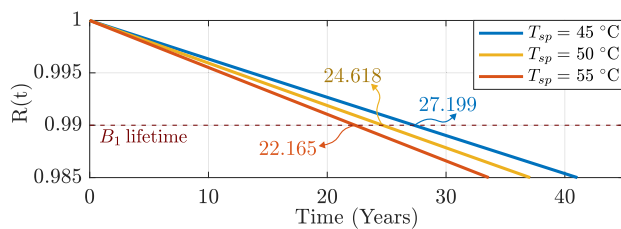


FIGURE 21. Effect of $\pm 5^\circ\text{C}$ variation in cooling fluid set-point (T_{sp}) on the B_1 lifetime of AFEC transistors.

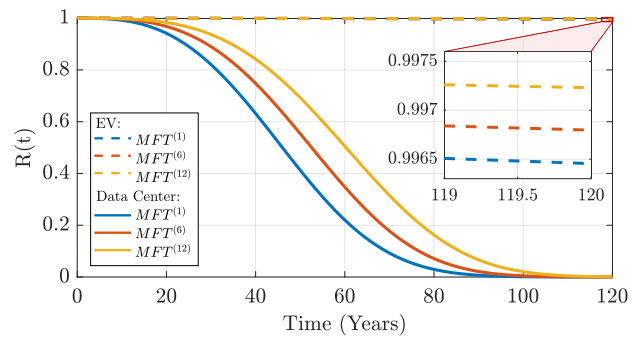


FIGURE 22. MFT reliability of different modules within one phase.

that this variation can cause a deviation of approximately ± 2.5 years in the B_1 lifetime of the AFEC transistors. The B_1 lifetime indicates 1% population failure translating to 1% probability of device failure occurring at a set operational period. This finding underscores the significant impact that each design aspect can have on the overall reliability of the system.

The SST control strategy distributes power equally among the modules, resulting in uniform stress on transistors, diodes, and capacitors across the modules. However, the difference in the MFT dielectric losses incur different stresses on the MFTs inside the topology. Fig. 22 shows the MFT reliability under the different mission profiles, emphasizing the impact of high voltage insulation on its lifetime. While the dielectric losses are only a small portion of the MFT losses, as presented in Table 6, the combination of the dielectric losses with the core and winding losses at high power lead to a more significant difference in reliability as depicted with the data center load.

Fig. 23 illustrates the impact of MFT reliability on module performance. Under EV load, the MFT has the lowest failure risk, minimally affecting module reliability, with the transformers in each module experiencing similar energy losses throughout the applied profile and therefore similar failure probabilities. This is reflected in the B_{10} lifetime data in Table 8. In contrast, under data center load, the MFT

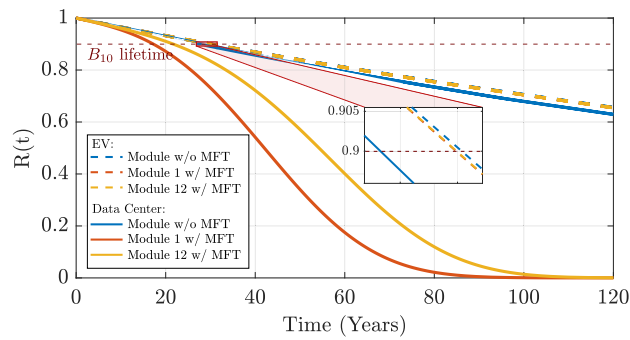


FIGURE 23. Module reliability w/ and w/o MFT consideration.

TABLE 8. Module B_{10} lifetime w/ and w/o MFT consideration.

Load Type	Module	B_{10} Lifetime
EV	Module w/o MFT	30.150 years
	Module 1 w/ MFT	29.965 years
	Module 12 w/ MFT	29.978 years
Data Center	Module w/o MFT	27.311 years
	Module 1 w/ MFT	17.202 years
	Module 12 w/ MFT	21.604 years

significantly increases the failure risk for each module. The results indicate that Module 1 reaches a 10% probability of failure ≈ 4.4 years earlier than Module 12. When the MFT is not considered, the results indicate that it would take ≈ 1.6 times longer for the module to reach the same level of unreliability as Module 1 with MFT.

Following the assessment of the individual components and modules, the overall reliability of the SST, which consists of 36 modules, is evaluated. Fig. 24 displays the SST reliability curve for two load profiles, both with and without considering the MFT reliability. Each module contains 35 components, resulting in a total of 1,260 components within the SST. This aggregation significantly increases the risk of failure, as illustrated by the B_{10} and B_{50} lifetimes shown in Fig. 24. These metrics indicate the probability of a single failure event among the 1,260 components, arising from a combination of random and wear-out failures.

Using (3), the MTTF of the four SST reliability curves are plotted in Fig. 25, highlighting the difference in the EV-load lifetime compared to that of data center load. This study identifies the components most likely to fail, specifically highlighting the designed MFT as the component most prone to failure in the context of the investigated data center load. Including the MFT in the reliability analysis results in a more than 15% reduction in MTTF for the data center load. It is important to note that these results are contingent upon the specific MFT design and may vary with minor modifications. For example, altering the wire insulation material or the cooling mechanism could improve the MFT's ability to withstand high operating temperatures and ultimately reduce its impact on system lifetime.

While the reference MFT is designed according to required system specifications and insulation distances, it still shows potential for degradation based on the utilized mission

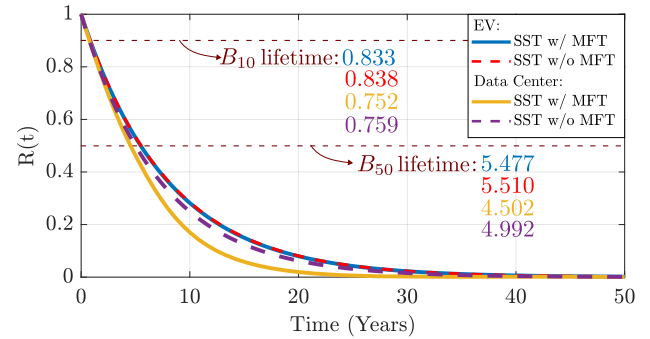


FIGURE 24. SST reliability w/ and w/o MFT consideration.

profiles. Additionally, the PDIV calculation in this work assumes that a mean void size is undefined; therefore, a range of potential void sizes is established to evaluate the likelihood of PD. If PD is deemed possible, the void length is set to the one achieving the highest $V_{void}/PDIV$, thereby accounting for the worst-case scenario. Based on the proposed MFT reliability framework, the effects of the electrical stress on the insulation lifetime model can be mitigated during the design phase by ensuring that the voltage potential across voids within the expected length range does not exceed the breakdown voltage of the void gas. On the other hand, this can also be addressed by determining the expected average void size from the encapsulation process and ensuring that the operating voltage remains below the breakdown voltage, in which case deviations in void size would be modeled using a normal distribution in the Monte Carlo analysis.

Furthermore, the difference in the SST MTTF between the two mission profiles, in the absence of MFT consideration, is only about one year. This similarity is attributed to the components all operating well below their maximum design limits, which minimizes degradation over time. As a result, the lifetimes in both scenarios remain closely aligned and decrease due to the multiple components within the system. The MTTF for the evaluated SST scenarios does not account for redundancy or scheduled maintenance, both of which could significantly improve system reliability and extend expected lifetime. Incorporating these factors suggests that the SST system has considerable potential to match the lifetimes of conventional architectures while benefiting from the added functionalities inherent in SSTs.

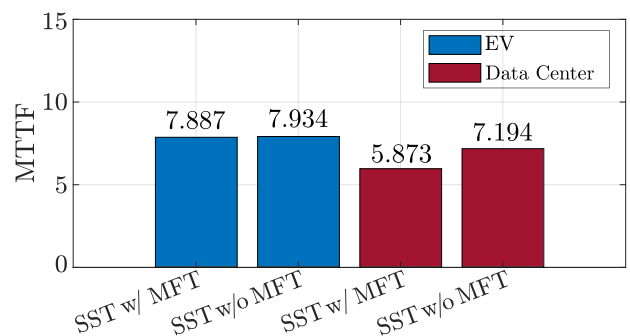


FIGURE 25. SST MTTF w/ and w/o MFT consideration.

VII. CONCLUSION

This article presents a comprehensive framework for designing and estimating the reliability of solid-state transformers. It integrates a proposed approach for medium frequency transformer reliability with established assessments in power electronics. The framework addresses both random failures and wear-out modes to evaluate system lifetime, highlighting how operational conditions and mission profiles significantly impact overall reliability. The reliability calculation is demonstrated through two test cases: an electric vehicle charging load and a data center load. These scenarios show how carefully including reliability factors in the design can enhance the understanding of SST longevity and support effective risk management. The investigation into MFT reliability reveals its critical influence on SST lifetime, a factor often overlooked in existing literature. The results show that factoring in the medium frequency transformer in reliability evaluations may lead to a significant reduction in the estimated lifespan of solid-state transformers, emphasizing the necessity of incorporating it into the overall reliability analysis. This framework helps confirm design robustness, prevents unnecessary over-design of safety margins, and ultimately regulates additional costs. Future work should focus on refining lifetime models, especially regarding MFT reliability, due to the limited data in the magnetic components domain. Additionally, exploring the variations of the components' electric parameters due to degradation over time will further improve the accuracy of SST lifetime estimation.

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AHMED MELIGY received the B.Sc. degree in electrical engineering from American University of Sharjah, in 2018, and the M.Sc. degree in renewable energy engineering and management from Albert-Ludwigs-Universität Freiburg, in 2022. He is currently pursuing the Industrial Ph.D. degree in electrical engineering with Grenoble Electrical Engineering Laboratory (G2Elab), Grenoble INP, France, in collaboration with Schneider Electric, France. He is also a Power Electronics Research Engineer with Schneider Electric. His research interests include solid-state transformers, multi-level converters, reliability, and optimization.



RAFAEL COELHO-MEDEIROS received the M.Sc. degree in electrical engineering from the National Polytechnic Institute of Toulouse, in 2018, and the Ph.D. degree from Paris-Saclay University, in 2022, France. He is currently with Schneider Electric, as a Lead Power Electronics Research and Development Engineer, with research interests in grid-connected power electronics converters, multiphysics modeling, real-time simulation, and power-hardware in the loop testing.



ILKNUR COLAK (Senior Member, IEEE) received the M.Sc. and Ph.D. degrees in electrical engineering from Istanbul Technical University, Istanbul, Türkiye. In last 23 years, she worked in industry and research centers, such as ABB, Ansaldo Richerche, TÜBITAK, CERN, and Maschinenfabrik Reinhausen. Since January 2022, she has been with Schneider Electric and leading the Medium Voltage Converter activities. Her research interests include multilevel converter topologies, modulation schemes, transformerless concepts, high power resonant converters, insulation-coordination, EMC and grounding, reliability, and wave energy conversion systems.



SEDDIK BACHA (Senior Member, IEEE) was born in Ighram (Béjaïa-Bgayet), Algeria, in 1958. He received the Engineering and master's degrees from the National Polytechnic School of Algiers (ENPAIger), in 1982 and 1990, respectively, the Ph.D. and Habilitation to Direct Research degrees from the National Polytechnic Institute of Grenoble, in 1993 and 1998, respectively, and the Algerian State Doctorate degree.

He has previously held many scientific responsibilities, both locally as the Head of a research group on electrical systems and networks (80 people) and at the national level, where he served as the Deputy Director of a CNRS Research Group gathering 400 researchers in electrical engineering in France. He is currently a Professor with the University of Grenoble Alpes. He is also the President of the Scientific Council of the SuperGrid Energy Transition Institute and leads one of its scientific programs. In parallel, he is also a Researcher with the G2Elab Laboratory, Grenoble. His scientific interests include optimal management of energy flows and modeling and control of electrical systems. His research applications cover renewable energy integration, electric mobility, HVDC networks, and microgrids. In these fields, he has supervised more than 60 doctoral theses and co-authored over 500 contributions, including two educational books on electric vehicles and one on the modeling and control of EP converters.

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